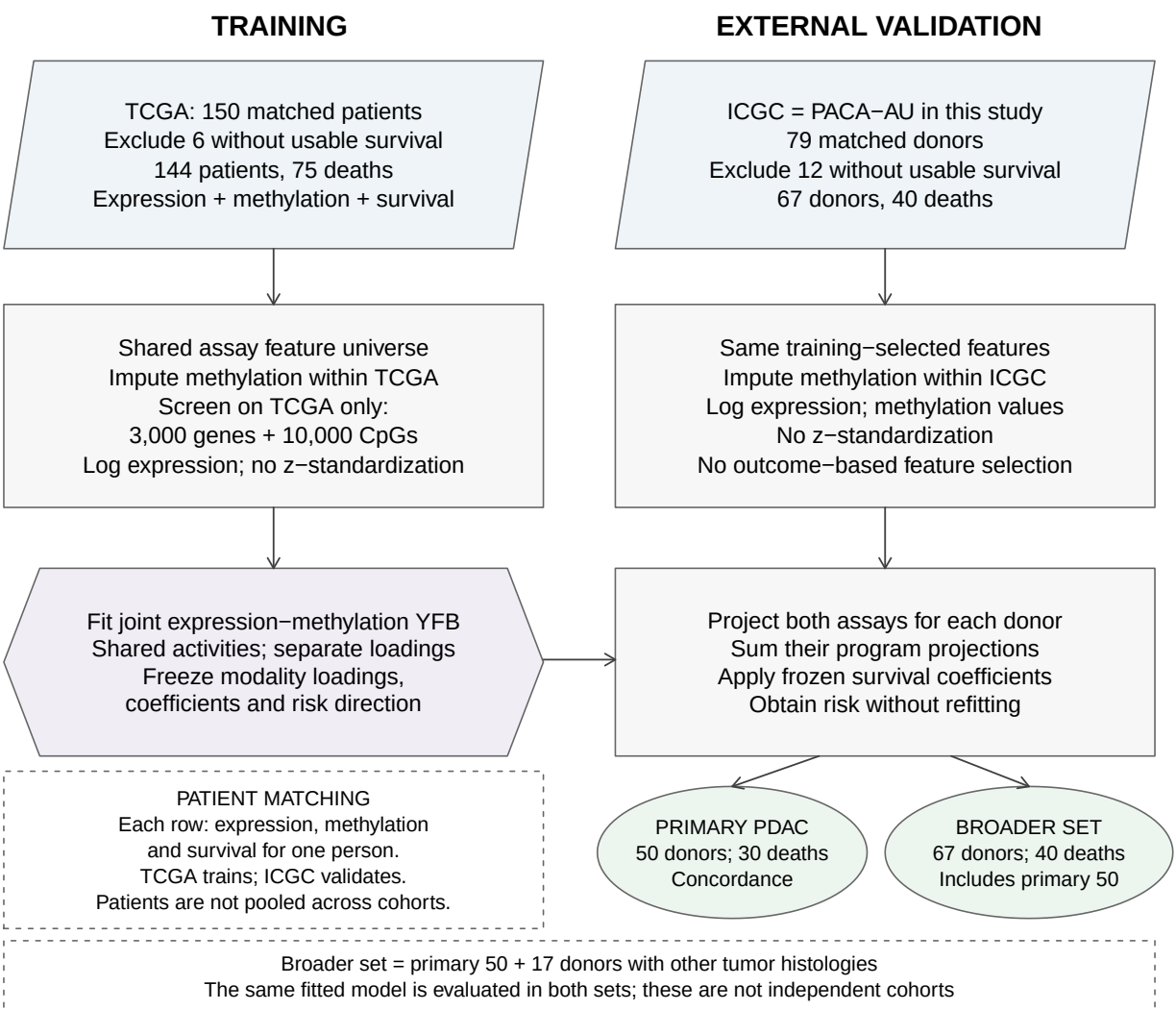
TRAINING
EXTERNAL VALIDATION
TCGA: 150 matched patients
Exclude 6 without usable survival
144 patients, 75 deaths
Expression + methylation + survival
ICGC = PACA−AU in this study
79 matched donors
Exclude 12 without usable survival
67 donors, 40 deaths
Shared assay feature universe
Impute methylation within TCGA
Screen on TCGA only:
3,000 genes + 10,000 CpGs
Log expression; no z−standardization
Same training−selected features
Impute methylation within ICGC
Log expression; methylation values
No z−standardization
No outcome−based feature selection
Fit joint expression−methylation YFB
Shared activities; separate loadings
Freeze modality loadings,
coefficients and risk direction
Project both assays for each donor
Sum their program projections
Apply frozen survival coefficients
Obtain risk without refitting
PATIENT MATCHING
Each row: expression, methylation
and survival for one person.
TCGA trains; ICGC validates.
Patients are not pooled across cohorts.
PRIMARY PDAC
50 donors; 30 deaths
Concordance
BROADER SET
67 donors; 40 deaths
Includes primary 50
Broader set = primary 50 + 17 donors with other tumor histologies
The same fitted model is evaluated in both sets; these are not independent cohorts